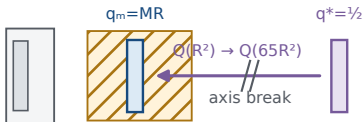


A Zero-mean shear / fixed centre

EXACT NSE

$$b(t, x_3) = B e^{-t} \cos x_3, \quad \int b \, dx_3 = 0$$



Version-A centre
 $x_c(t) = 0$ ($\bar{u} = 0$)

$A_m(R)$
 $\frac{2}{3}MR \leq |x| < \frac{4}{3}MR$

$$u = (AF, B e^{-t} \cos x_3, 0), \quad p = 0$$

$$F_0 = R^3 \partial_2 K^{\text{per}}(R^2; x_2 - q_{\text{pre}}) K^{\text{per}}(R^2; x_3)$$

scale-R packet; invariant in x_1

PROVED

ZERO GLOBAL MEAN

B Signed bridge / leakage

PROVED

$$d(t, x_3) = B e^{-t} (1 - \cos x_3) \leq 0$$

$$D = \int d(t-s, X_s) ds \leq 0$$

TARGET BRIDGE
 $E[|D| K^{\text{per}}(R^2; X_t)] \leq C R$

$$z < 0 \Rightarrow |z + D| \geq |z|$$

central half-chart; no periodic wrap

Exact exponential comparison

$$\text{target: } \exp(-M^2/288)$$

$$\text{leakage: } \exp(-M^2/264) \times \Pi(M)$$

$$1/264 - 1/288 = 1/3168 > 0$$

ALL COPIES

NO DNS

C Ratios / exact boundary

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$$R = \exp(-M^2/96), \quad A = R^{-2} \exp(M^2/576)$$

background
 $L / \text{row} = M \rightarrow \infty$

local leakage
 $L / \text{row} = [M/\Pi(M)]$
 $\times \exp[M^2(1/264 - 1/288)] \rightarrow \infty$

exterior residence
 $L / \text{row} = M^{7/3} \exp(M^2/288) \rightarrow \infty$

$$X_R / P_R^{2/3} \rightarrow \infty$$

**OPEN: local/mollified frame
or explicit entrance-flux payment**

2D3C / shear / moving-frame mechanisms: PRIOR ART
No priority claim; the frozen ledger result is narrow.

OPEN

PRIOR ART

NOT CLAY

Exact smooth periodic NSE • analytic source formulas only • fixed global-mean frame fails • no simulation